STATS 201A Lecture 7

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$$y_{ij} = \eta + \tau_i + \alpha_j + \varepsilon_{ij}, \quad i = 1, 2, 3, 4$$

$t^2 = F$

$$y = \beta_0 + \beta_1 X_A + \beta_2 X_B + \beta_3 X_A X_B + \beta_4 X_{blk} + \varepsilon$$

$$X = \begin{pmatrix} 1 & -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \quad X^\intercal X = \begin{pmatrix} 4 & & & \\ & 4 & & \\ & & 4 & \\ & & & 4 \end{pmatrix}$$

$$B_1 B_2 = (x_1 x_2)(x_1 x_3) = x_1^2 x_2 x_3 = x_2 x_3 = B_3$$

2^5 design, 32 parameters, 5 main effects, $\binom{5}{2}$ 2 way int, etc.